Datasets and Models

Which data trained what, why it was chosen, and what it measured

SIH 260486 • Image-based animal type classification for cattle and buffaloes
Team ASTRAL • August 22, 2026

How to read this document

Every figure here was produced by a command that can be re-run against this repository. Where something could not be established, it says so rather than being filled in plausibly.

Two numbers appear repeatedly and mean different things:

- **Held-out test accuracy** — measured on images the model never trained on, but drawn from the *same source* as its training data. This flatters a livestock model, because photographs from one farm share a backdrop, a camera and a lighting condition.
- **Source-held-out accuracy** — measured on photographs from a source the model has never seen at all. This is the number that predicts field behaviour, and it is the number we quote.

The gap between them is the whole reason this project reports what it reports. An earlier bovine breed model in this project scored **97.9%** on a held-out test set and was found to be classifying the *farm*, not the animal. Every model below is therefore reported against a control.

1 Summary

Model	Trained on	Status	Headline measure
RT-DETRv2 animal + ear tag	Indian cattle/buffalo detection set, 2 classes	Shipped	Ear tag read from decoder layer 1
HRNet-W32 41-joint keypoints	bovine41 (4,736 images)	Shipped, **22 of 41 joints**	PCK@0.05 89.9%, val PCK 0.9014
DINOv2 breed / group	Indian Bovine Breeds (41 classes, skewed)	Group head shipped, exact-breed head **disabled**	**80.2%** group source-held-out vs 60.7% control
SAM2 hiera-tiny segmentation	Not trained here — Meta, Apache-2.0	Shipped as published	Used for silhouette, weight, chest landmarks
DINOv2 + MLP skin lesion	Mendeley LSD 940 + Roboflow 163	**Not integrated**	0.882 held-out, **0.390** source-held-out

2 The dataset inventory

Counted on disk, not taken from folder names. Every Roboflow set below carries **CC BY 4.0**, declared in its own `data.yaml` or `README.dataset.txt`.

Dataset	Images	Licence	Used by
clean/bovine41_kaggle	4,736	CC BY 4.0	Keypoints — the merged set
clean/breed_merged	4,125	CC0-1.0 (export)	Breed / group probes
clean/breed_kaggle	4,125	CC0-1.0 (export)	Packaging copy of the above
clean/breed	2,348	CC BY 4.0	**The control that failed**
raw/luke/Indian_bovine_breeds	5,927	**Not established**	Breed source 1
raw/cattle-side-pose	2,576	CC BY 4.0	Keypoints (1,108 instances)
raw/bcs-hooks-pins	2,123	CC BY 4.0	Keypoints — **only hooks/pins source**
raw/livestock-keypoints	2,385	CC BY 4.0 (data)	Keypoints (9 of 18 joints map)
raw/cow-side-view	1,771	CC BY 4.0	Keypoints (1,297 instances)
raw/catbuf-indian-breeds	1,600	CC BY 4.0	Breed source 2 (789 kept)
raw/awa-pose	**2**	MIT (code)	**Never used — images not downloaded**
raw/cornell-teat-videos	**0**	—	**Empty. See below.**

Two entries that explain more than the rest of the table

`cornell-teat-videos` is an **empty directory**. It was the intended source of the udder and teat landmarks. Its emptiness is the direct cause of the nineteen untrained joints, and therefore of twelve traits that cannot score. The single most valuable thing anyone could add to this project is the contents of that folder.

`awa-pose` contributed **zero instances**. The annotation repository was cloned but the AwA2 images were never downloaded, so it appears in no source tag in `train.json` or `val.json`. It is listed here so nobody counts it as training data.

Two licences that are not established

Stated plainly because a judge may ask. The Kaggle *Indian Bovine Breeds* set (`luke`) carries **no licence file, no licence field, and no licence statement anywhere on disk**. The Mendeley LSD and Roboflow `cattle-smwai` sets behind the un-integrated lesion model are likewise unrecorded here. Everything else is CC BY 4.0, Apache-2.0, MIT or CC0.

3 Model 1 — RT-DETRv2, animal and ear-tag detection

Job. Find the animal in the frame, and find the ear tag. Everything downstream depends on it: no detection means no pose, no segmentation, and no measurement.

The version pin that matters more than it looks

`ml/requirements.txt` pins `transformers==5.0.0` exactly, and `server/preflight.py` *fails* if anything else is installed.

RT-DETRv2 stores its per-layer prediction heads **tied**. Version 5.15.1 expects them untied and fills the gap with random weights. The ear-tag path reads decoder layer 1 — one of the layers it randomises.

It does not crash. It returns a **confident-looking 0.62 detection with a box outside the image**, and the value changes on every process start. A silent wrong answer is far more expensive than a loud failure, which is why the pin is enforced by a preflight check rather than a comment.

Evidence: `models/README.md`, `ml/requirements.txt`, `server/preflight.py`

Why decoder layer 1

The final layer's ear-tag head collapsed to **AP@0.5 = 0.001**. Layer 1 is read instead. This is recorded in the manifest so that nobody re-discovers it by debugging a detector that appears to work for animals and not for tags.

Evidence: `models/MANIFEST.json`

4 Model 2 — HRNet-W32 bovine keypoints

Job. Place the anatomical landmarks that every conformation measurement is built from. **This model is the system's bottleneck, and the honest ceiling on what the product can score.**

Training

Run	`bovine41-pose-v2-hrnet384` (Kaggle)
Architecture	HRNet-W32 at 384 resolution
Checkpoint	epoch 42
Validation	val PCK 0.9014, PCK@0.05 89.9%
Schema	41 canonical joints
Joints trained	**22 of 41**

The nineteen joints nobody annotated

Measured over 40 real photographs, every one of these returns confidence **0.0%** — not low confidence, *zero*, on every image:

> `udder_floor`, `fore_udder_top`, `fore_udder_body_junction`, `rear_udder`, `rear_udder_top`, `rear_udder_left`, `rear_udder_right`, `udder_cleft_top`, `udder_cleft_bottom`, `teat_front_left`, `teat_front_right`, `teat_front_left_top`, `teat_front_left_bottom`, `teat_rear_left`, `teat_rear_right`, `teat_width_left_1`, `teat_width_left_2`, `vulva_base`, and the `rib_angle_*` triple.

The model was never taught these landmarks exist. **That blocks 12 of the 20 contract traits outright**, and it is a labelling task, not a modelling one.

What the model does place well

`chest_front`	100%	`withers`	70%
`hoof_left/right`	95%	`chest_bottom`	60%
`pastern_left`	95%	`shoulder_left`	55%
`tail_head`	90%	`hook_left`	38%
`hock_left`	88%	`pin_right`	**20%**

The pattern is consistent: the model learned the animal's hindquarters and legs well and its front and rump poorly. That is why leg-angle traits score and chest and rump traits mostly refuse.

The confidence threshold was measured, not chosen

`POSE_MIN_KEYPOINT_CONFIDENCE = 0.10`. Re-gating 55 photographs:

Gate	Usable joints	Measurements	In-band rate
0.30	6.5	41	34.1%
0.25	7.5	46	39.1%
0.20	8.7	62	35.5%
0.15	10.8	86	37.2%
0.10	**14.2**	**142**	**35.9%**
0.05	15.3	167	34.1%

The in-band rate is the *control*. If a looser gate were admitting misplaced joints, the extra measurements would scatter and that rate would fall toward chance. It does not move — so the measurements a strict gate was discarding were as plausible as the ones it kept, and there were three times as many. 0.05 is where the rate finally slips, so the knee at 0.10 is taken.

Evidence: `ml/config/models.py`

5 Model 3 — DINOv2 breed and group verification

Job. Check the photograph against the registered breed. The checkpoint is 81 KB because only the linear probes are stored; the DINOv2 backbone is fetched by `timm` on first use.

The exact-breed head disables itself

Measured at **38.1% source-held-out**. Its confidence carries no usable signal either: tightening the threshold from answering every time to answering 30% of the time moves accuracy by only 5.6 points.

The trained checkpoint therefore **switches its own breed head off**. The app never names a breed from a photograph, and `breed_verified: false` means **NOT CHECKED** — never "contradicted". Rendering it as a mismatch would accuse every correctly registered animal in the district.

The group head, and the control that makes it meaningful

Group head, source-held-out	**80.2%**
Background-only control	60.7%
Species head, source-held-out	88.5%

The background-only control is the important column. It measures what a model scores when it can see *only the backdrop* and not the animal. A model that scored 80% against a 79% control would have learned the farm. At 80.2% against 60.7%, this one has learned something about the animal.

Five groups: `red_zebu`, `grey_draught`, `dwarf_cattle`, `exotic_dairy`, `buffalo`. Where a group's own recall is poor — `exotic_dairy` at 43% — the response carries `group_reliable: false` and the app shows it as a hint rather than a finding.

Dataset skew

The Indian Bovine Breeds source has 36–439 images per class across 41 breed folders. That imbalance is the reason the exact-breed head cannot be trusted and the coarser group head can.

6 Model 4 — SAM2 hiera-tiny

Not trained by this team. Meta's published checkpoint, Apache-2.0, used as-is for silhouette segmentation.

It matters more than its size suggests. The 2D-to-3D weight estimator, the `chest_bottom` landmark and the udder floor all depend on a real silhouette. SAM2 was once *absent from the server environment entirely*, and `segment_animal` silently fell back to the detection box as a plain rectangle on every image — disabling all three without a single error. It is now pinned in the requirements and checked at preflight.

Evidence: `STATUS-measured.md`, `ml/detection/detector.py`

7 Model 5 — skin lesion detector (not integrated)

Status: not in the shipped system. No checkpoint is in this repository and none is on any branch. This is why the app reports **NOT SCREENED** rather than a clean bill of health — an empty symptom list is not a clinical finding.

Source	Images	Labels
Mendeley LSD	940	`skin_nodules`: 289 positive, 651 negative
Roboflow `cattle-smwai`	163	`circular_skin_patches`: 99 positive, 64 negative
Total	1,103	cluster-split 769 / 165 / 169

Architecture: DINOv2 ViT-S/14 frozen, MLP head 384 → 128 → 4, masked BCE so that −1 labels contribute zero gradient (verified by gradient inspection, not asserted).

Skin nodules	Held-out test	Source-held-out
F1	0.882	**0.390**
Precision	0.911	0.471
Recall	0.854	0.333

Why it is not integrated

Two blockers, neither about the model itself.

The source-held-out figure may be measuring an untrained head. Each symptom's labels live entirely in one source — Mendeley carries `skin_nodules` and masks circular patches, Roboflow the reverse. Holding out a source therefore removes *all* supervision for that head. A true source-held-out experiment may not be possible on this dataset as built, and the fix would be skin-nodule images from a second source.

The background-only control did not complete. It returned empty metrics after RT-DETR checkpoint warnings. A run that errored and predicted nothing yields the same 0.000 recall as a model that correctly refused, so it cannot be read as the control passing.

By this project's own standard — the exact-breed head disables itself at 38.1% — `skin_nodules` at 0.390 should also disable itself. `circular_skin_patches` at 0.889 sits nearer the group head's 80.2%, but rests on only 9 positive test examples (Wilson 95% CI 0.57–0.98).

8 What this means for the product

The trait ceiling is not a bug and not a tuning problem:

Traits	Count	Blocked by
Blocked	12	Udder, teat and rib landmarks never annotated
Weak	3	`pin_right` resolves 20% of the time
Reachable	5	Of which 5 need a centimetre scale from the ear tag

So the honest ceiling today is **5 to 8 traits**, and the scorecard states which ones and why each of the rest refused. Reaching 20 requires an annotation session, not a better model.

Model weights are excluded from git and fetched by `scripts/fetch_models.py`, which verifies a SHA256 against `models/MANIFEST.json` and refuses a mismatch — a truncated download otherwise behaves exactly like a broken model.